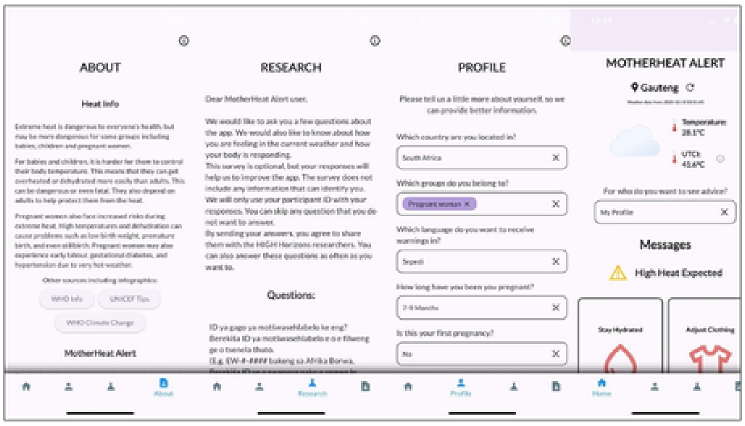
BLOG POST

RETHINKING DIGITAL CLIMATE INNOVATION FOR MOTHERS LIVING THROUGH EXTREME HEAT

Across the world, extreme heat is quietly becoming one of the most dangerous climate threats to human health. Yet one group remains largely invisible in many heat-health warning systems: pregnant and postpartum women and their infants, whose bodies are least able to cope with extreme heat. Pregnancy already places additional strain on the body. Add rising temperatures, prolonged heat exposure and limited access to cooling or healthcare, and the risks multiply; from dehydration and heat stress to complications affecting both mother and child. Despite this, most existing early warning systems are designed for the general populations, overlooking the unique vulnerabilities of mothers.

The Heat Indicators of Global Health (HIGH Horizons) project, set out to close this critical gap by developing MotherHeat Alert, a smartphone-based heat early warning system designed specifically for pregnant and postpartum women. Implemented in Sweden, South Africa, and Zimbabwe, the app was tested across diverse settings from occasional heatwaves in high-income communities to persistent heat exposure in low-resource environments. Using local temperature data and heat stress indicators, MotherHeat Alert sends timely warnings alongside practical, easy-to-follow advice to help women protect themselves and their babies. Importantly, researchers worked directly with women through participatory approaches, including photovoice and early warning system messaging, where participants documented their lived experiences, through photographs, with the aim of understanding how heat affected their daily lives and what advice would actually be useful. The result was guidance that reflected local realities rather than generic public health messaging.



MotherHeat Alert App Interface

While the project spanned three countries with vastly different climate and health systems, the lessons emerging from South Africa offer a powerful reminder that successful climate innovation is about much more than technology.

What Happened in South Africa?

In South Africa, MotherHeat Alert was implemented among 200 pregnant and postpartum women receiving antenatal and postnatal care in Mamelodi, Tshwane. Research nurses supported participants with app installation during clinic visits, creating trust and ensuring that users understood how the system worked from the beginning. That human touch proved essential. The app worked, but not always in the way researchers expected. Once women returned home, everyday realities began to shape their experience with the technology.

Participant with baby, phone, umbrella on health facility bench

Limited mobile data, insufficient phone storage, older devices, delayed notifications, and unreliable electricity reduced the app's performance for some users. These were reminders that climate solutions operate within people's daily lives, not under ideal conditions. Researchers also discovered another challenge that not software update alone could fix.

Beyond Technology: The Human Factor

Receiving a heat alert does not automatically translate into protective action. Researchers found that many women needed additional support to understand:

- Why heat poses health risks during pregnancy
- What the alerts actually mean and why they mattered
- How to respond effectively
- Why the recommendations are important

Health literacy, risk perception and ongoing support proved just as important as the technology itself. The lesson was clear: a digital intervention can only be effective as the system surrounding it.

What Policymakers should take away

The HIGH Horizons team offers lessons that extend far beyond maternal health.

- Design for context, not for convenience. Heat warning systems are socio-technical interventions, not simply digital products. Delivery must adapt to local infrastructure and health systems.
- People matter as much as platforms, community health workers, nurses, frontline health professionals and other trusted intermediaries play a critical role in helping communities understand, trust and consistently use climate-health innovations.
- Embed early warning systems into routine care such as antenatal and postnatal care, health education sessions and community health worker interactions. This increases credibility, reinforces messages and makes climate adaptation part of everyday healthcare rather than a standalone intervention.

As global temperatures continue to rise, protecting mothers and babies from heat can't be an afterthought. MotherHeat Alert shows that tailored, co-designed warning systems are possible, but they succeed only when technology is woven into the health services and human relationships people already trust.

Case study: Heat Indicators of Global Health (HIGH Horizons): Monitoring, Early Warning Systems and Health Facility Interventions for Pregnant and Postpartum Women, Infants, Young Children and Health Workers.
Shared as part of the CAPCHA case study collection by Pascalia Munyewende on behalf of the HIGH Horizons Study Group, Wits RHI Implementation Science Climate and Health (HIGH Horizons)

Compiled by the CAPCHA Team: Ann Irungu, Virginia Wamboi and Maria Nailantei